

QUESTION :

Sample Java program to check whether the given number is prime palindrome or not.

EXECUTION STEPS :

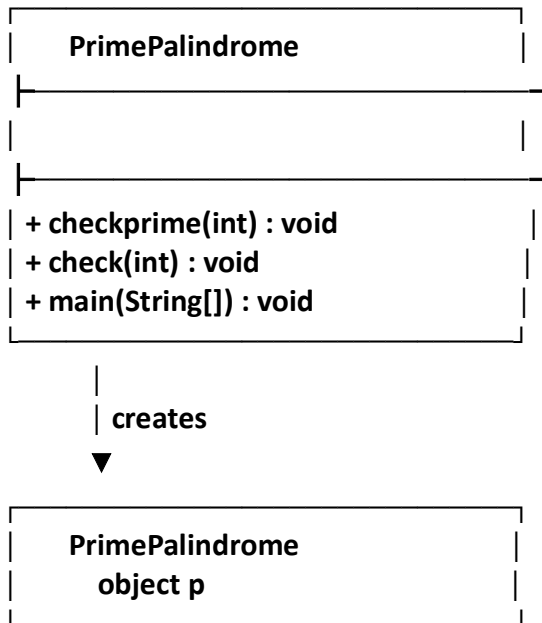
- **Create the Java project**
 - Open IntelliJ IDEA.
 - Create a new Java project and select the required JDK.
- **Create the class**
 - Create a Java class named `PrimePalindrome`.
 - Make sure the file name is `PrimePalindrome.java`.
- **Enter the program**
 - Copy and paste the given Java code into the class.
 - Save the program using **Ctrl + S**.
- **Run the program**
 - Click the green **Run** button near `main()`.
 - Enter a number in the console when prompted.
- **View the output**
 - The program first checks whether the number is prime.
 - If it is prime, it checks whether the number is a palindrome.
 - The final result is displayed in the IntelliJ console.

ALGORITHM :

1. **Start** the program.
2. Read a number `num` from the user using `Scanner`.
3. Check whether the number is **0 or negative**.
 - If yes, display **"Invalid"**.
 - Otherwise, continue.
4. Check whether the number is **1**.
 - If yes, display **"The number is not a Prime Palindrome"** and stop.
5. Check whether the number is **prime**:
 - Set `x = true`.
 - Start a loop from `i = 2` to `num - 1`.
 - If `num % i == 0`, the number is not prime.
 - Set `x = false` and stop the loop.
6. If `x` is `true`, check whether the number is a **palindrome**.
7. To find the reverse:
 - Store the original number in `real`.
 - Set `rev = 0`.
 - Extract the last digit using `n % 10`.
 - Remove the last digit using `n / 10`.
 - Construct the reverse using `rev = rev * 10 + r`.
8. Compare `rev` with `real`.
 - If `rev == real`, display **"The given number is a prime palindrome"**.
 - Otherwise, display **"The number is prime but not a palindrome"**.

9. **Stop** the program.

UML DIAGRAM :



CODE :

```
import java.util.Scanner;

public class PrimePalindrome {
    void checkprime(int num) {
        boolean x=true;
        if(num<0 || num==0) {
            System.out.println("the check for ZER or negative INVALID");
        }
        else if(num==1) {
            System.out.println("The number is not a Prime Palindrome");
            return;
        }

        else {
            for(int i=2;i<num;i++) {
                if(num%i==0) {
                    System.out.println("the number is not a primepalindrome");
                    x=false;
                }
            }
        }
    }
}
```

```

                break;
            }
        }

        }
        if(x) {
            check(num);
        }
    }
    void check(int n) {
        int real=n;
        int rev=0,r;
        while(n>0) {
            r=n%10;
            n=n/10;
            rev=rev*10+r;
        }
        if(rev==real) {
            System.out.println("the given number is a primepalindrome");
        }
        else {
            System.out.println("tehe number is prime but not a palindrome");
        }
    }
}
public static void main(String[] args) {
    int number;
    Scanner sc=new Scanner(System.in);
    System.out.println("enter the number you wanna check for a primepalindrome");
    number =sc.nextInt();
    PrimePalindrome p=new PrimePalindrome();
    p.checkprime(number);
}
}

```

OUTPUT :

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=53548" -Dfile.encoding=UTF-8 -Dsun
enter the number you wanna check for a primepalindrome
34
the number is not a primepalindrome

Process finished with exit code 0
|

```